

stm32_devboard_routed

Design Report

kicad-tools 0.13.0

Rev 1 | 2026-05-13 | jlcpcb

Board Summary

Property	Value
Layers	2 copper (F.Cu, B.Cu)
Footprints	17 (0 SMD, 0 THT, 17 other)
Nets	12
Traces	223 segments
Vias	26
Board Size	60.0 x 40.0 mm

Design Overview

Theory of Operation

STM32F103C8 Development Board

End-to-end design example

Demonstrates circuit blocks API

Power Architecture

Power Rails: +3V3, +5V, GND

Regulator	Device
U1	AMS1117-3.3

Assembly Notes

1 fine-pitch component; 1 polarized component

- **Fine-pitch components:** 1 (U2)
- **Polarized components:** 1 – check orientation markings

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: SKIPPED – ERC skipped by user request

Bill of Materials

Value	Package	Qty	References
100nF	C_0805_2012Metric	5	C3, C12, C13, C14, C15
10uF	C_0805_2012Metric	2	C1, C2
20pF	C_0805_2012Metric	2	C10, C11
4.7uF	C_0805_2012Metric	1	C16
USER	LED_0805_2012Metric	1	D1
SWD-6	PinHeader_1x06_P2.54mm_Veritical	1	J1
10k	R_0805_2012Metric	1	R2
330R	R_0805_2012Metric	1	R1
AMS1117-3.3	SOT-223-3_TabPin2	1	U1
STM32F103C8T6	LQFP-48_7x7mm_P0.5mm	1	U2
8MHz	Crystal_HC49-4H_Veritical	1	Y1

DRC Status

Metric	Count
Errors	101
Warnings	0
Blocking	101

Status: FAIL ### Violations by Type

Violation Type	Count
clearance_pad_segment	44
dimension_via_drill	14
dimension_via_diameter	14
dimension_annular_ring	14
clearance_segment_via	4
clearance_pad_via	4
clearance_segment_segment	4
via_in_pad	3

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- **[CRITICAL]** Fix 101 blocking DRC violations (clearance_pad_segment (44), dimension_via_drill (14), dimension_via_diameter (14))
- **[CRITICAL]** Increase min via drill: 0.200mm < 0.300mm required
- **[OPTIONAL]** Verify zone fill in KiCad for 3 zone-connected nets

Routing Status

Metric	Value
Signal Net Completion	100.0% (9/9)
Overall Completion	91.7%
Complete Nets	11 / 12
Zone-Connected Nets	3
Incomplete Nets	1
Unconnected Pads	3

Zone-Connected Nets

- +3.3V
- +5V
- GND

Incomplete Nets

- GND

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~0.88 USD
Components (estimated)	~1.36 USD
Assembly (estimated)	~2.02 USD
Total (estimated)	~4.26 USD
Batch Quantity	5
Batch Total (estimated)	~21.3 USD